Discrete Mathematics (PCC-CS401)

Course Details

Credit Points: 4

Pre-Requisites

1. Some concepts from basic math – algebra, geometry, pre-calculus syllabus outline

Unit 1

- Sets, Relation and Function: Operations and Laws of Sets, Cartesian Products, Binary Relation, Partial Ordering Relation, Equivalence Relation, Image of a Set, Sum and Product of Functions, Bijective functions, Inverse and Composite Function, Size of a Set, Finite and infinite Sets, Countable and uncountable Sets, Cantor's diagonal argument and The Power Set theorem, Schroeder-Bernstein theorem.
- Principles of Mathematical Induction: The Well-Ordering Principle, Recursive definition.
- The Division algorithm: Prime Numbers
- The Greatest Common Divisor: Euclidean Algorithm, The Fundamental Theorem of Arithmetic.

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Unit 2

- Basic counting techniques-inclusion and exclusion, pigeon-hole principle, permutation and combination

Unit 3

- Propositional Logic: Syntax, Semantics, Validity and Satisfiability, Basic Connectives and Truth Tables.
- Logical Equivalence: The Laws of Logic, Logical Implication, Rules of Inference, The use of Quantifiers.

- Proof Techniques: Some Terminology, Proof Methods and Strategies, Forward Proof, Proof by Contradiction, Proof by Contraposition, Proof of Necessity and Sufficiency.

Unit 4

Algebraic Structures and Morphism

- Algebraic Structures with one Binary Operation, Semi Groups, Monoids, Groups, Congruence Relation and Quotient Structures, Free and Cyclic Monoids and Groups, Permutation Groups, Substructures, Normal Subgroups, Algebraic Structures with two Binary Operation, Rings, Integral Domain and Fields.
- Boolean Algebra and Boolean Ring, Identities of Boolean Algebra, Duality, Representation of Boolean Function, Disjunctive and Conjunctive Normal Form

Unit 5

Graphs and Trees

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- Graphs and their properties, Degree, Connectivity, Path, Cycle, Sub Graph, Isomorphism, Eulerian and Hamiltonian Walks, Graph Colouring, Colouring maps and Planar Graphs, Colouring Vertices, Colouring Edges, List Colouring, Perfect Graph, definition properties and Example, rooted trees, trees and sorting, weighted trees and prefix codes, Bi-connected component and Articulation Points, Shortest distances.